

OXFORD

INTERNATIONAL
AQA EXAMINATIONS

INTERNATIONAL AS CHEMISTRY

(9620) CH02

Report on the examination

January 2024

REPORT ON EXAMINATION: INTERNATIONAL AS CHEMISTRY (9620) CH02 JANUARY 2024

Students generally answered the questions well and are good at Equilibrium calculations.

Some points to note to improve candidate outcomes in future examinations.

- Writing balanced equations needs to be developed in students' study of Chemistry.
- Questions need to be read carefully to extract the relevant information.
- Students' have learnt organic reagents well, but recall of other facts, e.g. catalyst names, should be improved.
- Names of and reasons why practical techniques are used needs to be known.

QUESTION 01

1.1 Most students scored this mark. It was surprising that several students could not write this equation even though all compounds were given in question.

1.2 Most students were able to calculate M_r , but not as many were able to convert this to molecular formula. C_7H_{16} was a common error.

1.3 Reading the question carefully would improve answers here. The question was about unburned hydrocarbons and not pollution by other compounds.

1.4 Most students correctly stated acid rain. An unexpected number said it caused global warming or damaged ozone.

1.5 This was answered well.

1.6 Approximately two thirds of students answered well. The area under the curve means the total area under curve not the proportion that have $E \geq E_a$. Error was to think that the number of molecules and area changed.

1.7 Students know how a catalyst increases the rate. The best students discussed frequency of collisions or collisions per unit time.

1.8 Not as well-known as expected. Silver was a common error.

QUESTION 02

2.1 This was challenging for students. Most answers described how to separate hydrocarbons based on their different boiling points instead of defining what a fraction is. It must be similar not the same boiling point. Another error was stating it was cracking.

2.2 The question was answered well but marks were lost by not naming and stating where van der Waals forces occur. The answer must be comparative so should state weaker not weak.

2.3 Answered well with approximately two thirds of students scoring full marks. A common error was drawing displayed or cyclic structures or dimethyl propane which is given in question 02.2

2.4 Common error to say product is more valuable or in demand which is a reason to do cracking but not why the atom economy is 100%

QUESTION 03

3.1 Polymer repeating units was answered well. Common mistake was putting 3 carbons in a straight line in the repeating unit.

3.2 Not as many students scored this mark as expected. The common mistakes were students discussing intermolecular forces or not having $C=C$.

3.3 This proved to be difficult for students. The name needs to be buta not but, diene not ene.

QUESTION 04

4.1 About two thirds of students scored full marks. Cyanide ion must have negative charge and lone pair on the carbon of CN .

4.2 This was answered well with students stating $C-I$ bond being weaker.

4.3 Equation was answered well but students must remember all products have to be present in an equation and not have ions only on one side of the arrow.

QUESTION 05

5.1 The question discriminated well with only the best students scoring full marks. Errors included using an incorrect primary carbocation and a carbocation is $+$ not $\delta+$. The bromide ion is Br^- not $\delta-$.

5.2 Some good explanations were seen; marks were lost when students did not discuss the number of alkyl chains present on the carbocation.

5.3 Free radical steps were attempted well. The role of uv light is to break $Br-Br$ bond to form Br radicals, not act as a catalyst.

QUESTION 06

6.1 Molecular formula is C_3H_8O not C_3H_7OH . Workings should be shown using the ratio of moles of each element. Credit was given on this occasion if the candidate stated the correct formula with alternative workings.

6.2 Displayed formula was surprisingly not answered well. The common error was missing $O-H$ bond in displayed formula.

6.3 Students' knowledge of organic reagents was answered well throughout the paper. But an error was to try to identify the compound by giving a molecular formula.

6.4 This was answered well. The common error was not stating acidified in the reagent.

6.5 This was difficult for students to score full marks. Aldehyde must be CHO not COH and water is needed as the other product in the equation. Numbering in naming of aldehydes needs to start from the aldehyde functional group.

6.6 The bond angle for epoxyethane was well known but the structure of the reaction product was more difficult. A methyl ether and diol were common errors.

QUESTION 07

7.1 Students found drawing the mechanism for a cyclic structure difficult. A common error was to do base catalysed elimination steps even though the question stated acidic conditions.

7.2 This was well known, although there were a surprising number of blank responses.

7.3 The naming of practical techniques was unexpectedly difficult for students. There were several blanks and incorrect answers.

7.4 Students found this question difficult. Purity of liquids is checked by having a narrow range (sharp) boiling point that matches known data. Credit was given for discussing density and comparison with known data.

7.5 This calculation was answered well by approximately two thirds of students. Errors were to use the wrong data from the table and question.

QUESTION 08

8.1 The equilibrium expression was answered well but students need to use square not round brackets. The units were not always worked out correctly from the expression.

8.2 Students were good at calculating the equilibrium quantities.

8.3 Students were able to explain equilibrium well. To improve, students must explain why equilibrium moves. A surprising error was for students to discuss rates of reaction.

8.4 This was not answered as well as expected. Equal effect on forward and reverse reactions by a catalyst is needed.

8.5 The majority of students were able to score this mark.

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